

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

Step 2: Create Expense Dataset

```
data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Food': [5000, 5500, 6000, 5800, 6200, 6500],
    'Transport': [2000, 2200, 2500, 2300, 2700, 3000],
    'Shopping': [3000, 3500, 3200, 4000, 4500, 5000],
    'Bills': [1500, 1500, 1600, 1700, 1800, 1900]
}
```

```
df = pd.DataFrame(data)
```

```
df
```

Step 3: Data Cleaning

```
print("Missing Values")
```

```
print(df.isnull().sum())
```

```
df.drop_duplicates(inplace=True)
```

```
print("Data Cleaning Completed")
```

Step 4: Total Monthly Expense

```
df['Total_Expense'] = (
    df['Food']
    + df['Transport']
    + df['Shopping']
    + df['Bills']
)
```

```
df
```

Step 5: Expense Trend Visualization

```
plt.figure(figsize=(8,5))
```

```
plt.plot(
    df['Month'],
    df['Total_Expense'],
    marker='o'
)
```

```
plt.title("Monthly Expense Trend")
```

```
plt.xlabel("Month")
```

```

plt.ylabel("Expense")

plt.grid(True)

plt.show()
Step 6: Category-wise Expense Analysis
categories = [
    'Food',
    'Transport',
    'Shopping',
    'Bills'
]

totals = [
    df['Food'].sum(),
    df['Transport'].sum(),
    df['Shopping'].sum(),
    df['Bills'].sum()
]

plt.figure(figsize=(8,5))

plt.bar(categories, totals)

plt.title("Expense Category Analysis")
plt.xlabel("Category")
plt.ylabel("Amount")

plt.show()

```

7: Expense Distribution Pie Chart

```

plt.figure(figsize=(6,6))

plt.pie(
    totals,
    labels=categories,
    autopct='%1.1f%%'
)

plt.title("Expense Distribution")

plt.show()

```

Step 8: Analysis Commands

```

print("Total Expense =", df['Total_Expense'].sum())

print("Average Monthly Expense =", df['Total_Expense'].mean())

print("Highest Expense =", df['Total_Expense'].max())

```

```
print("Lowest Expense =", df['Total_Expense'].min())
```

Step 9: Prediction Model

Predict next month's expense.

```
X = np.array([1,2,3,4,5,6]).reshape(-1,1)
```

```
y = np.array(df['Total_Expense'])
```

```
model = LinearRegression()
```

```
model.fit(X,y)
```

```
prediction = model.predict([[7]])
```

```
print("Predicted Expense for Next Month =", prediction[0])
```

Step 10: Forecast Graph

```
future_months = [1,2,3,4,5,6,7]
```

```
future_expense = list(y)
```

```
future_expense.append(prediction[0])
```

```
plt.figure(figsize=(8,5))
```

```
plt.plot(
    future_months,
    future_expense,
    marker='o'
)
```

```
plt.title("Expense Forecast")
```

```
plt.xlabel("Month Number")
```

```
plt.ylabel("Expense")
```

```
plt.grid(True)
```

```
plt.show()
```

Step 11: Dashboard Creation

```
fig, axs = plt.subplots(2,2, figsize=(12,8))
```

```
axs[0,0].plot(df['Month'], df['Total_Expense'])
```

```
axs[0,0].set_title('Total Expense')

axs[0,1].bar(df['Month'], df['Food'])
axs[0,1].set_title('Food')

axs[1,0].bar(df['Month'], df['Shopping'])
axs[1,0].set_title('Shopping')

axs[1,1].bar(df['Month'], df['Transport'])
axs[1,1].set_title('Transport')

plt.tight_layout()

plt.show()
```